



Independent Sideband Transmission (ISB)

and

Vestigial Sideband Transmission (VSB)

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Independent Sideband (ISB)

- Form of AM in which a single carrier frequency is independently modulated by two different modulating signals.
- One modulator produces USB and one modulator produces LSB.
- ISB conserves both transmit power and bandwidth.
- Used in United states for stereo AM transmission.

Independent Sideband (ISB)

- **Bandwidth Efficiency:**

- Reduces bandwidth usage compared to traditional AM
- Two sidebands are transmitted separately
- Each sideband carries different information

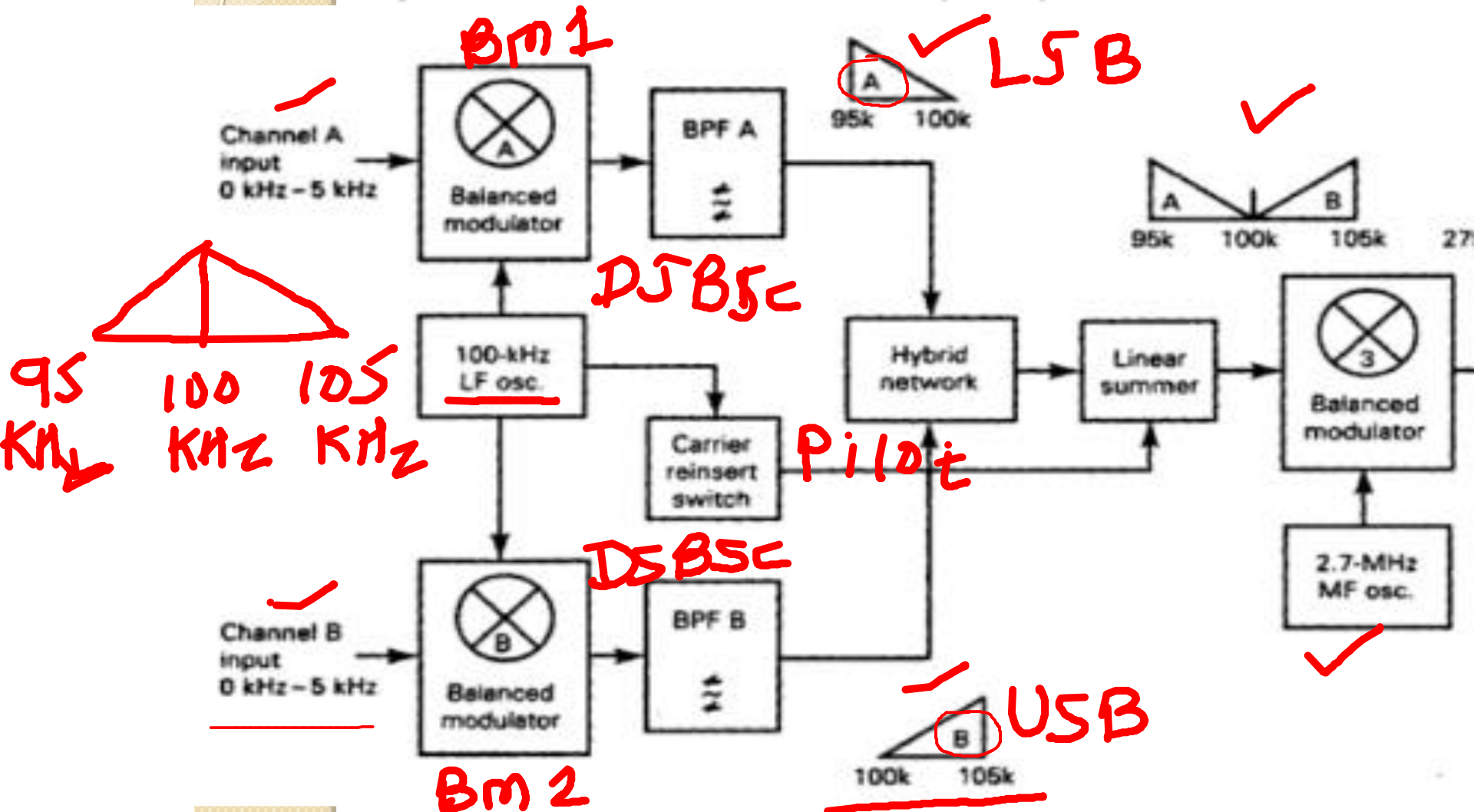
- **Power Efficiency:**

- Power is not wasted by transmitting same information in both sidebands (like DSBFC)
- Results in more efficient use of the transmitted power

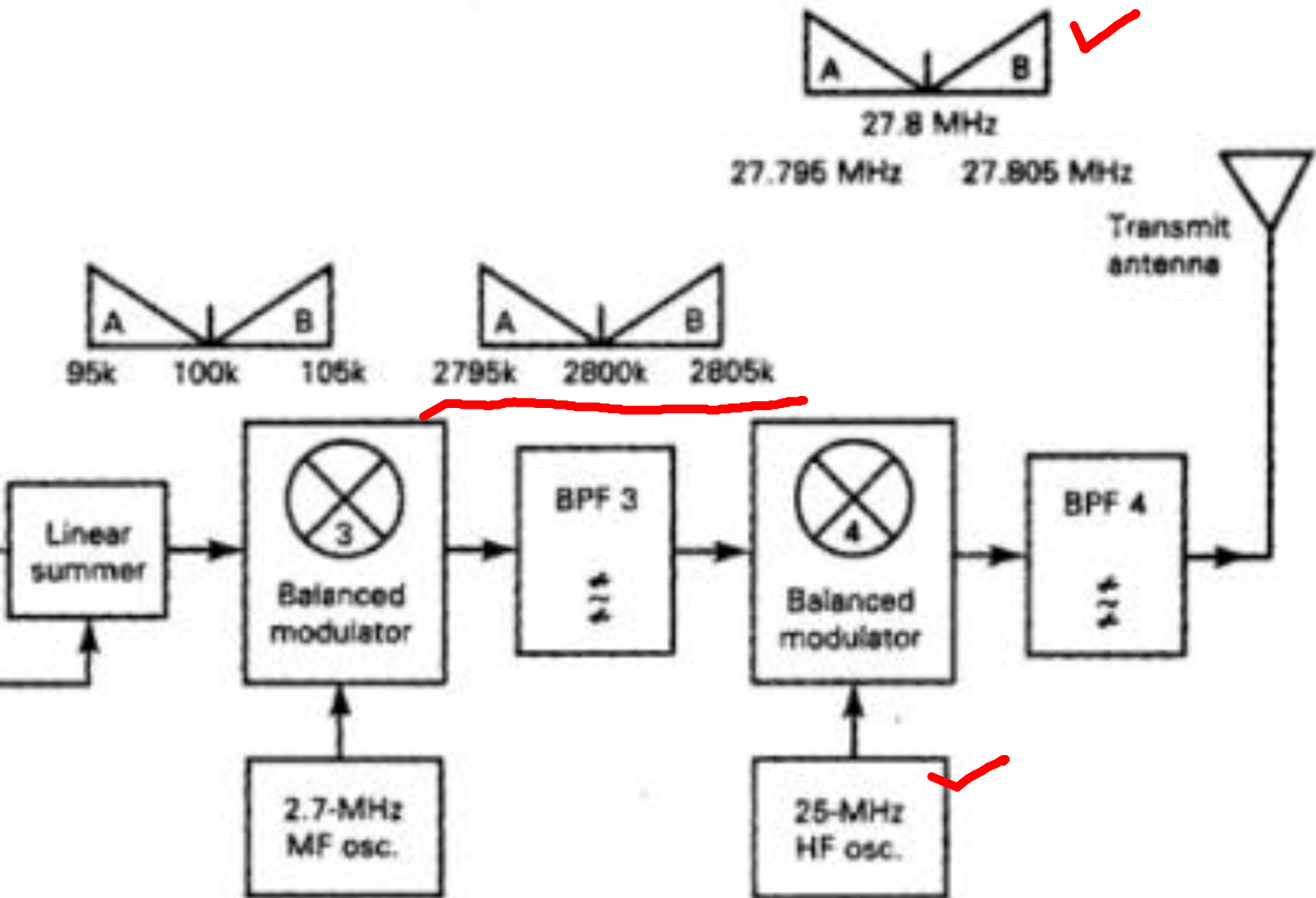
Independent Sideband (ISB)- Applications

- **Radio Communication**
 - Multiple signal transmitted over the same channel
 - Some military and amateur radio applications
- **Broadcasting:**
 - Used in some high-fidelity audio transmissions and satellite communications

Independent Sideband (ISB) Transmitter



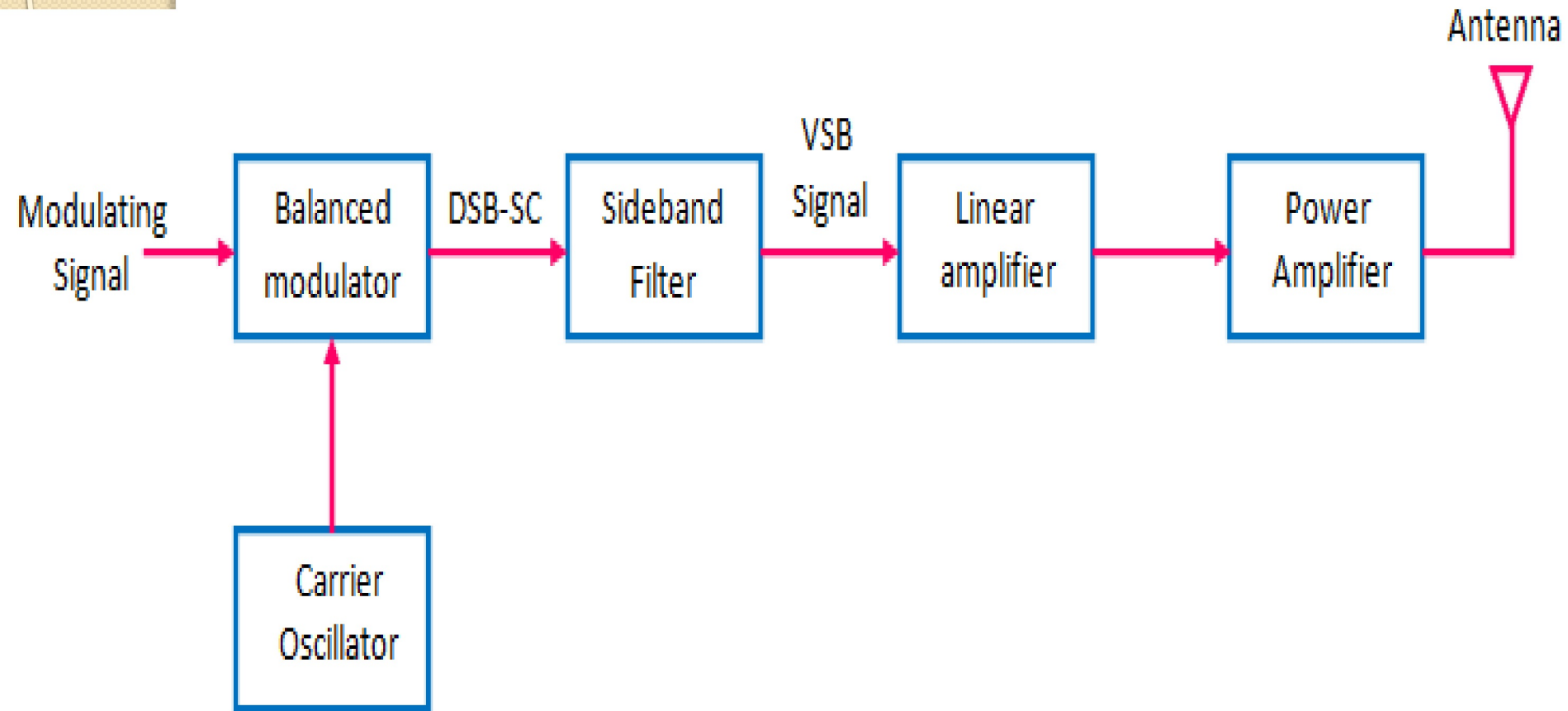
Independent Sideband (ISB) Transmitter



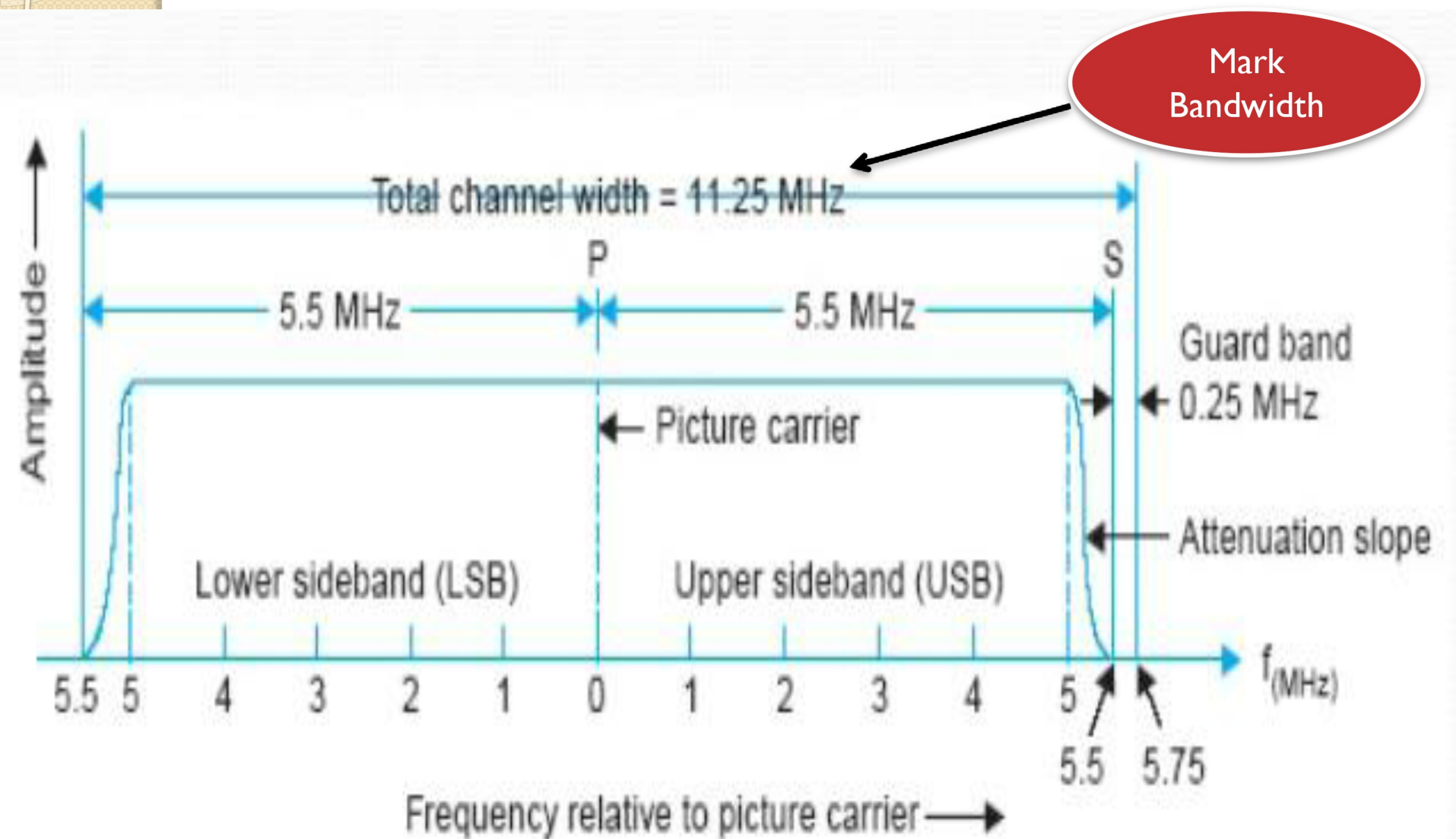
Vestigial Sideband (VSB)

- VSB is similar to **SSB** but it retains a small portion (a vestige) of the **unwanted** sideband.
- VSB signals are generated using standard AM or DSB-SC modulation, then passing modulated signal through a **sideband shaping filter**.
- VSB modulation with envelope detection are used to modulate image (video) in **analog TV signals**.

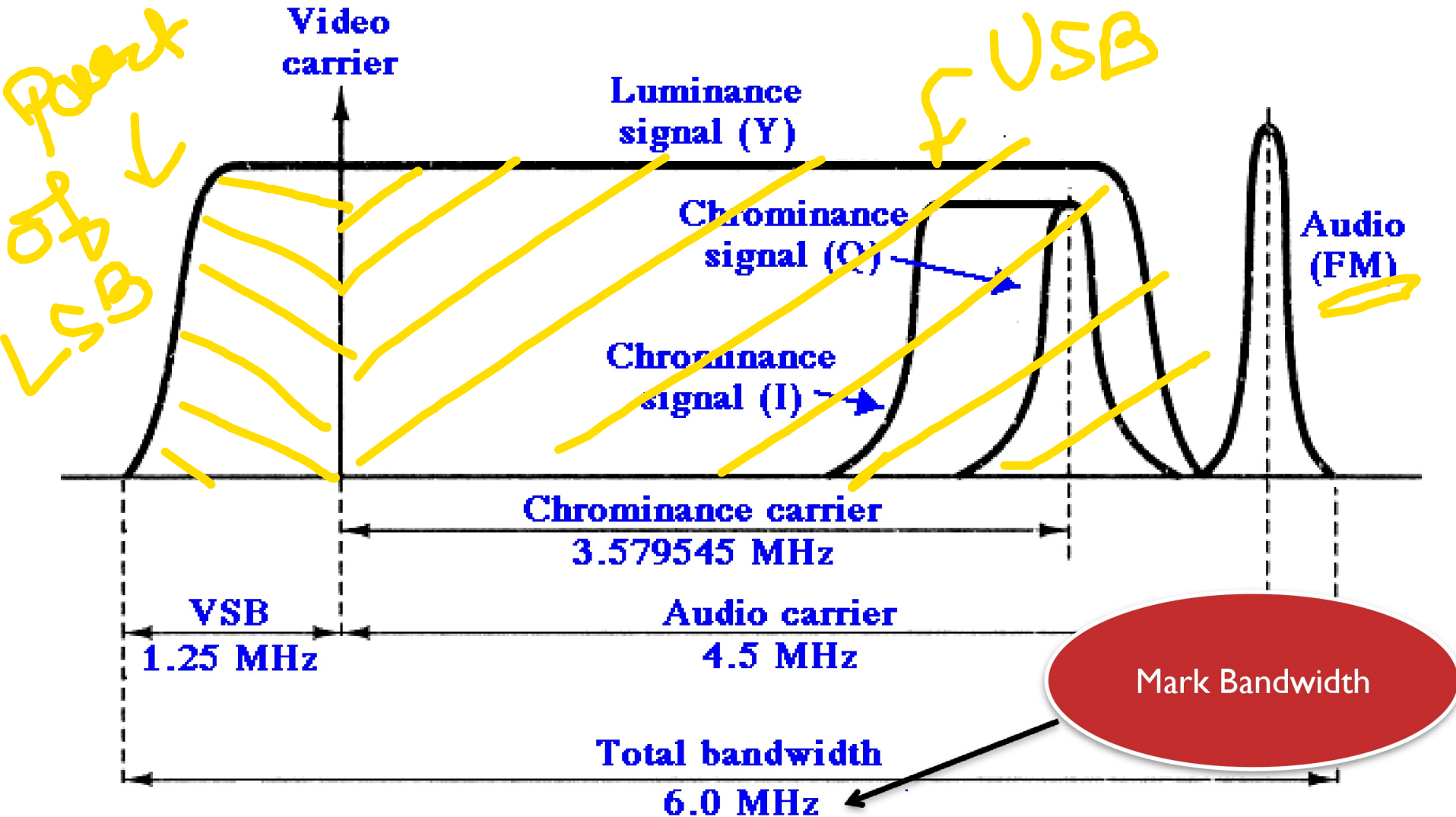
Vestigial Sideband (VSB) Transmitter



Vestigial Sideband (VSB) Concept



Vestigial Sideband (VSB) Concept



To Summarize AM Systems

S.No.	Parameter of comparison	DSB FC (Standard AM)	DSBSC	SSB	VSB
1.	Carrier suppression	N.A.	Fully	Fully	N.A.
2.	Sideband suppression	N.A.	N.A.	One S.B. completely	One S.B. suppressed partially
3.	Bandwidth	$2f_m$	$2f_m$	f_m	$f_m < BW < 2f_m$
4.	Transmission efficiency	Minimum	Moderate	Maximum	Moderate
5.	No. of modulating inputs	1	1	1	2
6.	Application	Radio broadcasting	Radio broadcasting	Point to point mobile communication	T.V.